Ceiling Diffuser FD



Description
Aluminum rectangular diffuser with individual adjustable blades.

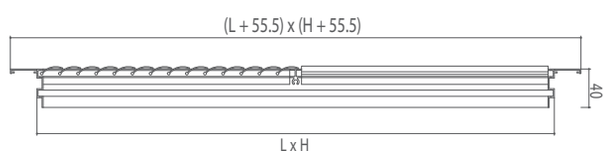
Application
Designed for air exhaust from ventilation and air conditioning systems.

Characteristics
The diffuser is equipped with adjustable blades arranged in four directions, causing rotation of the air with Coanda effect, obtaining a high rate of induction and reduced stratification.
On request the diffuser can be provided with 1, 2 or 3 discharge directions.
Recommended for rooms with high ceilings (eg. factories, airports, sports halls, etc.)
Minimum recommended height is 3.8 m.
Designed specifically for integration into the paneled ceiling box (dimensions 600 x 600 mm), the speaker is provided without mounting holes.
On request, the diffuser can be produced in custom sizes, in which case it is provided with mounting holes.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
With screws, visible on the diffuser's frame.

Selection table FD

Air flow (m^3/h)	Dimensions	595x595
	Ak [m^2]	0.1819
960	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.5 3.0 23.0 13.0
1150	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.8 3.8 30.0 18.0
1380	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.1 4.5 35.0 25.0
1610	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.5 5.5 38.0 33.0
1860	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.8 6.2 42.0 47.0
2300	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	3.4 7.5 50.0 75.0
2700	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	4.1 8.0 55.0 100.0

Annotation

* Ak [m^2] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Length air jet at speed Vk = 0.2 m/s
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Note: Table values are for maximum blade opening

Order codes

Type + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **FD**
- **ADP IZ/N**: Insulated/ Non-insulated adapter
- **Dimensions**: **L X H** — 595 x 595
- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints